DEPI · SYSTEM ANALYSIS & DESIGN · DELIVERABLE 2 OF 6

Database Design & Data Modeling

Section 4.2 — ER diagram, logical and physical schema

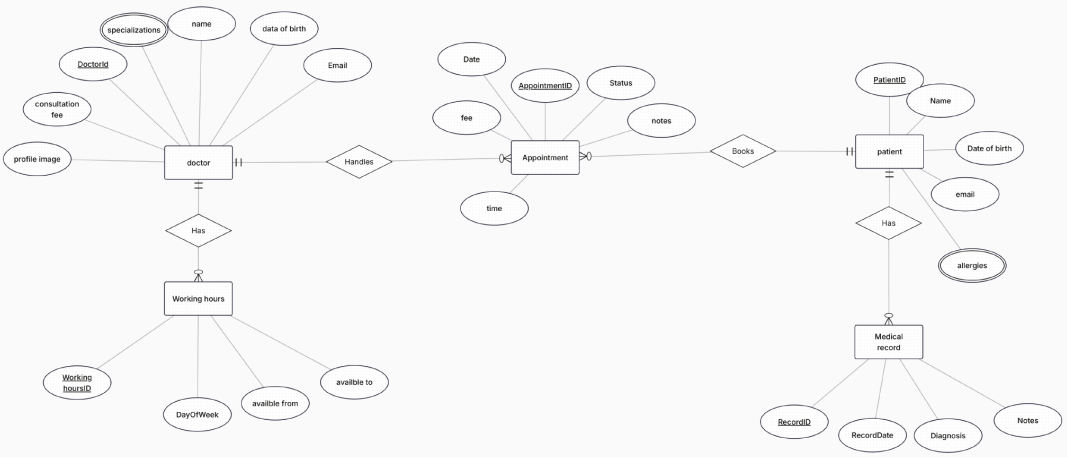
MediCare — Clinic Management & Appointment System

ASP.NET Core MVC · N-Tier · Microsoft Azure

Team: Mohamed Munir (Team Lead) · Malak Hany · Ashraf Ibrahim · Leqaa Ibrahim · Natalie Bola

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5. ER Diagram



Entities and attributes (as drawn)

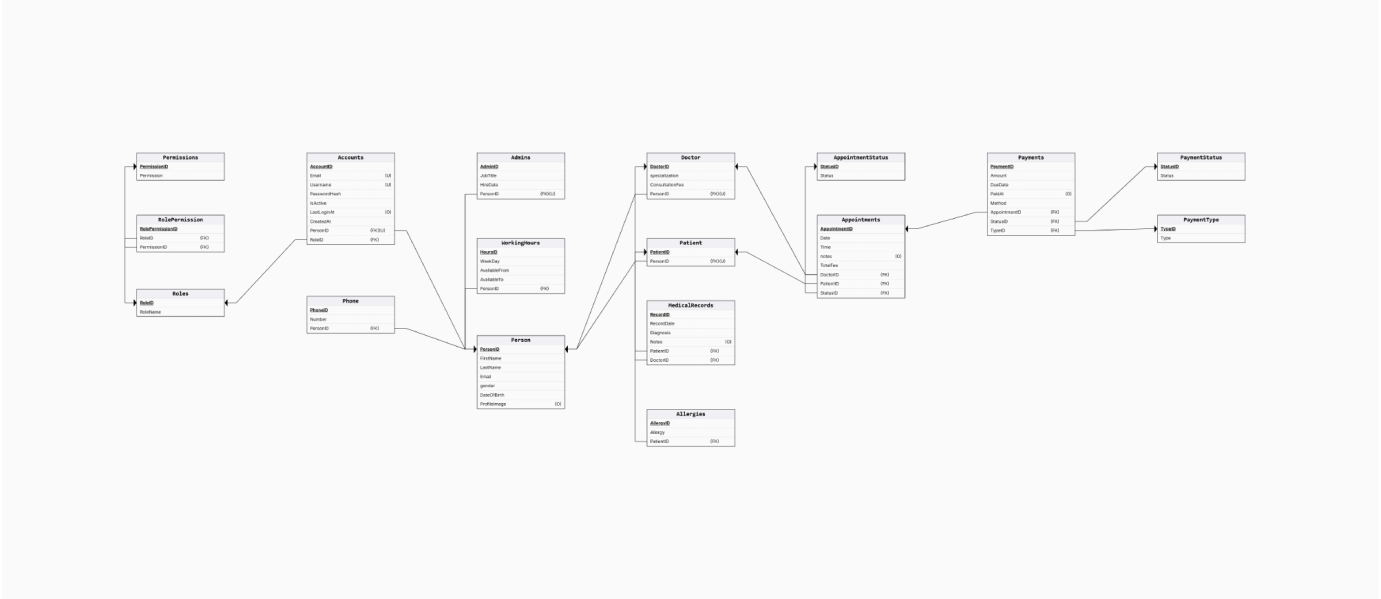
Entity	Attributes	Notes
Doctor	**DoctorId** (PK), name, date of birth, email, consultation fee, profile image, *specializations* (multi-valued)	
Working hours	**WorkingHoursID** (PK), DayOfWeek, available from, available to	
Appointment	**AppointmentID** (PK), date, time, status, fee, notes	
Patient	**PatientID** (PK), name, date of birth, email, *allergies* (multi-valued)	
Medical record	**RecordID** (PK), RecordDate, Diagnosis, Notes	

Relationships

Relationship	Between	Cardinality (as drawn)
Has	Doctor — Working hours	one doctor has zero-or-many working-hours rows
Handles	Doctor — Appointment	one doctor handles zero-or-many appointments
Books	Patient — Appointment	one patient books zero-or-many appointments
Has	Patient — Medical record	one patient has zero-or-many medical records

6. Logical & Physical Schema

Logical schema (relational)



Notation: **PK** primary key · FK foreign key · (U) unique · (O) optional/nullable.

Table	Columns
Permissions	**PermissionID** (PK), Permission
Roles	**RoleID** (PK), RoleName
RolePermission	**RolePermissionID** (PK), RoleID (FK), PermissionID (FK)
Accounts	**AccountID** (PK), Email (U), Username (U), PasswordHash, IsActive, LastLoginAt (O), CreatedAt, PersonID (FK, U), RoleID (FK)
Person	**PersonID** (PK), FirstName, LastName, Email, gender, DateOfBirth, ProfileImage (O)
Phone	**PhoneID** (PK), Number, PersonID (FK)
Admins	**AdminID** (PK), JobTitle, HireDate, PersonID (FK, U)
Doctor	**DoctorID** (PK), specialization, ConsultationFee, PersonID (FK, U)
Patient	**PatientID** (PK), PersonID (FK, U)
WorkingHours	**HoursID** (PK), WeekDay, AvailableFrom, AvailableTo, PersonID (FK)
MedicalRecords	**RecordID** (PK), RecordDate, Diagnosis, Notes (O), PatientID (FK), DoctorID (FK)
Allergies	**AllergyID** (PK), Allergy, PatientID (FK)
AppointmentStatus	**StatusID** (PK), Status
Appointments	**AppointmentID** (PK), Date, Time, notes (O), TotalFee, DoctorID (FK), PatientID (FK), StatusID (FK)
Payments	**PaymentID** (PK), Amount, DueDate, PaidAt (O), Method, AppointmentID (FK), StatusID (FK), TypeID (FK)
PaymentStatus	**StatusID** (PK), Status
PaymentType	**TypeID** (PK), Type

Normalization considerations

- **Person** is a shared supertype for Doctor, Patient and Admin (one row per human; each role table references it 1:1 via a unique `PersonID`).
- Multi-valued attributes from the ERD are split into tables: **Allergies** (per patient) and **Phone** (per person).

- Roles and permissions are modelled as **Roles** / **Permissions** / **RolePermission** (many-to-many), and statuses as lookup tables (**AppointmentStatus, PaymentStatus, PaymentType**).